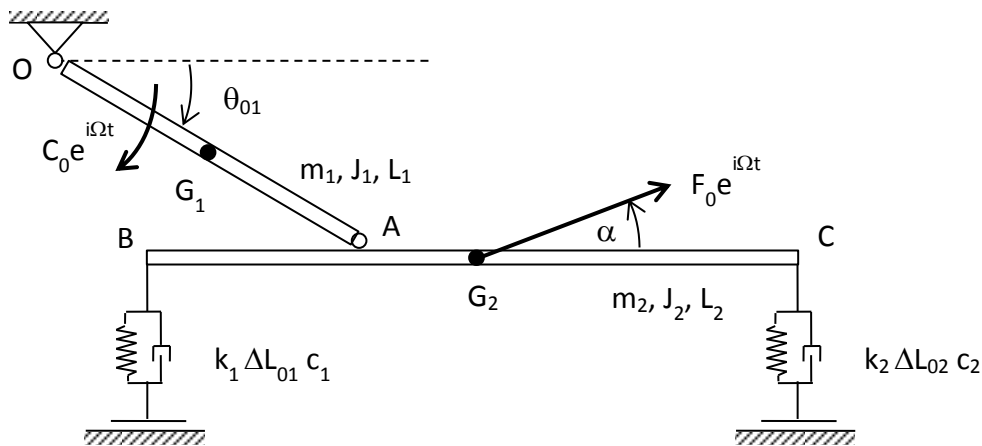


Mechanical Systems Dynamic

February 04th, 2016



$m_1 = 5 \text{ kg}$	$k_1 = 1000 \text{ N/m}$
$m_2 = 10 \text{ kg}$	$k_2 = 2000 \text{ N/m}$
$J_1 = 0.2 \text{ kgm}^2$	$\Delta L_{01} = -0.0654 \text{ m}$
$J_2 = 1.2 \text{ kgm}^2$	$\Delta L_{02} = -0.0286 \text{ m}$
$L_1 = 0.7 \text{ m}$	$c_1 = 2 \text{ Ns/m}$
$L_2 = 1.2 \text{ m}$	$c_2 = 0.2 \text{ Ns/m}$
$AB = 0.4 \text{ m}$	$\theta_{01} = \pi/6 \text{ rad}$
$AC = 0.8 \text{ m}$	$\alpha = \pi/6 \text{ rad}$

The mechanical system above lays in the vertical plane and is shown in its equilibrium position.

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the torque $C_0 e^{i\Omega t}$. Output: the absolute rotation of the BC bar and the horizontal component of the displacement of point G1. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the $F_0 e^{i\Omega t}$ inclined by a constant angle α with respect the horizontal direction. Output: the relative rotation between the BC bar and the OA bar, and the force transmitted by the spring and damper system ($k_2 c_2$). Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the torque $C_0 e^{i\Omega t}$. Output: the normal component to the OA bar of the constraint force in the hinge A Save the results as SNxxx5.
6. Compute the static preload of the springs k_1 and k_2 changing the position of the hinge A along the BC bar and keeping the angle θ_{01} equal to $\pi/6$ and the BC bar horizontal: in the new position of the hinge A the distances AB and AC are 0.8 m and 0.4 respectively. Write the numerical result on the sheets of the solution.

Note: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Please write also the SNxxx code on the first sheet of the solution, close to the surname, name and identification number.

Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test Feb. 04 2016*
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse...* (or *Sfoggia...*)
7. Fill the Title field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.